

POSITION PAPER

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THE ROLE OF IT PROFESSIONALS IN ENSURING ETHICAL AI PRACTICES

INTRODUCTION

Artificial Intelligence (AI) has become an integral part of modern technology, influencing various sectors such as healthcare, finance, education, and more. As AI systems continue to evolve and integrate into our daily lives, the importance of ethical practices in AI development and deployment has become paramount. IT professionals play a critical role in ensuring that AI systems are developed, implemented, and maintained ethically. This involves addressing issues related to bias, transparency, accountability, privacy, and security. This paper explores the responsibilities of IT professionals in fostering ethical AI practices, highlighting key challenges and proposing strategies for effective ethical management.

5 PRO-LITERATURE -EVIDENCES W/REFERENCES

Ethical Considerations in Artificial Intelligence: A Comprehensive Discussion from the Perspective of Computer Vision

To address these concerns, stakeholders including researchers, industry leaders, and policymakers are actively developing comprehensive frameworks and guidelines to promote responsible AI practices. This paper delves into the ethical dimensions of computer vision, highlighting key areas for improvement. It underscores the critical need to address biases and ensure equitable outcomes, protect individual rights, and maintain data security. Furthermore, it advocates for increased public engagement and transparency in AI development processes to build trust and accountability. The integration of ethical principles into the design and deployment of computer vision systems is crucial for navigating these challenges and fostering responsible innovation.

- **Diverse and Representative Datasets:** Key to minimizing algorithmic bias and ensuring equitable outcomes.

- **Informed Consent:** Essential for protecting individual rights and data security.
- **Robust Privacy Protocols:** Important for maintaining data security.
- **Public Engagement and Transparency:** Advocated to build trust and accountability in AI development processes.
- **Integration of Ethical Principles:** Crucial for responsible innovation in computer vision systems.

The Ethical Implications of Artificial Intelligence (AI) For Meaningful Work

In examining the interplay between AI and meaningful work, literature highlights several critical dimensions. It emphasizes that work perceived as meaningful is characterized by a comprehensive scope of tasks and the opportunity to utilize a diverse skill set. Additionally, it shows that meaningful work involves not only fulfilling tasks but also contributing to broader social and personal development. These dimensions provide a robust foundation for analyzing how AI might alter these aspects, with implications for the meaningfulness of work in increasingly automated environments.

- **Hackman and Oldham's Job Characteristics Model (1976):** Highlights the importance of task integrity and skill variety for meaningful work.
- **Empirical Research (Pratt & Ashforth, 2003):** Supports the idea that job features enhancing scope and skill variety improve workers' sense of purpose and engagement.
- **Lips-Wiersma and Morris's Holistic Framework (2009):** Integrates dimensions such as personal growth and service to others, showing that meaningful work involves broader social and personal development.
- **Philosophical Perspectives by Wolf (2010):** Argues that meaningful activities should be engaging, complex, and beneficial to others, encompassing both subjective and objective elements of meaningfulness.

PRINCIPLED ARTIFICIAL INTELLIGENCE: Mapping Consensus in Ethical and Rights-based Approaches to Principles for AI

The research identifies essential themes for ethical AI, including privacy, accountability, safety and security, transparency, fairness, human control of technology, professional responsibility, and promotion of human values. However, applying these high-level principles in practice remains challenging. There is a need for more detailed studies, especially concerning specific AI applications such as facial recognition and autonomous vehicles. The landscape of AI ethics is diverse and evolving, requiring further research to address variations across regions

and stakeholders. While existing legal frameworks and literature provide a foundation, regulations and litigation concerning AI are still emerging. Tracking the influence of ethical principles on AI-related liability will be crucial as technology continues to develop.

- **Eight Key Themes for Ethical AI:** Privacy, Accountability, Safety and Security, Transparency and Explainability, Fairness and Non-discrimination, Human Control of Technology, Professional Responsibility, and Promotion of Human Values.
- **Challenges in Bridging High-Level Principles to Real-World Application:** Emphasizes the difficulty in translating ethical principles into practical application.
- **Need for Detailed Studies:** Highlights the requirement for more research on specific AI applications like facial recognition and autonomous vehicles.
- **Diverse and Evolving AI Ethics Landscape:** Points to the ongoing variations and need for further research across different regions and stakeholders.
- **Existing Legal Frameworks and Academic Literature:** Provides a foundation but acknowledges that regulations and litigation are still developing.
- **Tracking Ethical Principles' Influence on AI Liability:** Stresses the importance of monitoring how ethical principles affect AI-related liability as technology evolves.

Ethical principles for artificial intelligence in education

The literature strongly supports the integration of ethical principles in the development and deployment of AI in education. The establishment of global standards for AI ethics underscores the significant role of ethics in AI. The transformative potential of AI in education is emphasized, highlighting how AI can enhance educational practices and foster new paradigms of instruction and technology advancement. Despite these benefits, ethical considerations are paramount. Ethical challenges including privacy concerns, data biases, and the need for transparency must be addressed to ensure trustworthy AI in education. This comprehensive analysis provides a robust framework for developing ethical AI, promoting a balanced approach to leveraging AI's potential while safeguarding human values and rights.

- **UNESCO (2021):** Establishment of global standards for AI ethics, signed by 193 member countries, highlighting the role of ethics in AI.
- **Becker et al. (2018) and Seldon with Abidoye (2018):** Emphasize the transformative potential of AI in education.
- **Hwang et al. (2020):** Illustrates AI's role in enhancing educational practices, acting as an intelligent tutor and policy-making advisor.

- **Holmes et al. (2021) and Kay & Kummerfeld (2019):** Discuss ethical challenges such as privacy concerns, data biases, and the need for transparency in AI in education.
- **Reports from UNESCO, OECD, and the European Commission:** Provide a comprehensive framework for developing ethical AI, promoting a balanced approach to leveraging AI while safeguarding human values and rights.

The Role of Systems Engineers in Ensuring Ethical AI Practices

The role of systems engineers in ensuring transparency and accountability in AI systems is supported by extensive literature highlighting their pivotal contributions to ethical AI. Studies underscore the necessity for transparent AI systems to prevent bias and discrimination, emphasizing the importance of explainable AI techniques to make AI decision-making processes understandable and interpretable. Literature also emphasizes the value of regular ethical audits and assessments, which help identify and rectify biases and errors in AI systems, thereby promoting fairness and accountability. The establishment of guidelines and standards for AI development provides a structured approach to integrating ethical principles into AI systems, ensuring they adhere to best practices and ethical norms. Continuous monitoring and improvement of AI systems are essential for maintaining transparency and accountability, with literature highlighting the importance of adaptive and proactive measures to address evolving biases and errors. By designing, auditing, and continuously improving AI systems, systems engineers play a crucial role in fostering trust and acceptance among users and stakeholders, contributing to the responsible deployment of AI technologies.

- **Doshi-Velez and Kim (2017):** Emphasize the importance of explainable AI techniques for making decision-making processes understandable and interpretable.
- **Binns (2018):** Highlights the value of regular ethical audits and assessments to identify and rectify biases and errors in AI systems.
- **Floridi et al. (2018):** Suggest the establishment of guidelines and standards for AI development to integrate ethical principles and ensure adherence to best practices.
- **Morley et al. (2019):** Stress the importance of continuous monitoring and improvement of AI systems to address evolving biases and errors.
- **Ananny and Crawford (2018):** Argue that systems engineers play a crucial role in designing, auditing, and improving AI systems to foster trust and acceptance.

5 CONS-LITERATURE

-EVIDENCES W/REFERENCES

Ethics in AI through the practitioner's view: a grounded theory literature review

The literature consistently underscores the critical importance of integrating ethical considerations into AI development. The swift adoption of AI across various sectors has necessitated a focus on ethical principles to govern AI behaviors. Case studies illustrate the severe consequences of neglecting ethics, such as backlash from the developer community and performance issues in AI technologies related to race and gender. These incidents underscore the urgent need for ethical frameworks and guidelines to ensure fairness, accountability, and transparency in AI applications. Furthermore, the necessity of incorporating practitioners' perspectives to effectively implement ethical principles in AI systems is emphasized. Collectively, these studies and examples highlight the pivotal role of ethics in AI development, advocating for rigorous ethical standards to mitigate risks and promote socially responsible AI.

- **Hagendorff (2020)**: The swift adoption of AI across various sectors has necessitated a focus on ethical principles to govern AI behaviors.
- **GitHub**: Use of unlicensed source code sparked significant backlash from the developer community (Al-Kaswan & Izadi, 2023).
- **Facial recognition technologies**: AI systems performed poorly on Black women compared to white men (Buolamwini & Gebru, 2018).
- **Amazon's AI recruitment tool**: The tool had to be discontinued due to gender bias (Dastin, 2018).
- **Khan et al. (2022) and Leikas et al. (2019)**: Emphasize the necessity of incorporating practitioners' perspectives to effectively implement ethical principles in AI systems.

The race for an artificial general intelligence: implications for public policy

The literature emphasizes the significant impact of AI on various sectors and underscores the necessity of ethical considerations to mitigate potential risks. AI is highlighted as one of the most profound technological advancements, even more critical than electricity or fire. This sentiment is reflected in numerous examples of AI applications in daily life, such as virtual digital assistants, automated vehicles, and advanced medical diagnostics. Despite these advancements, current AI technologies are limited to specific tasks and lack the general problem-solving abilities of Artificial General Intelligence (AGI). The potential future development of AGI presents both immense opportunities and significant risks, as it could lead to unprecedented economic growth and societal changes. However, the competitive race to develop AGI may result in compromised safety standards and ethical lapses, potentially leading to disastrous consequences. Therefore, there is an advocacy for stringent ethical frameworks and government policies to steer AI development towards positive and safe outcomes, emphasizing the need for public control, taxation, and coordinated efforts to ensure the creation of a "friendly" AGI that benefits humanity as a whole.

- **Sundar Pichai:** Highlights AI as one of the most profound technological advancements, more critical than electricity or fire.
- **Makridakis (2017):** Provides examples of AI applications in daily life, such as virtual digital assistants (Amazon's Alexa, Apple's Siri), automated vehicles, and advanced medical diagnostics.
- **LeCun et al. (2015):** Discusses that current AI technologies, known as narrow AI, are limited to specific tasks and lack the general problem-solving abilities of Artificial General Intelligence (AGI).
- **Kurzweil (2005); Nordhaus (2015):** Mention the potential future development of AGI, which could lead to significant economic growth and societal changes.
- **Bostrom (2017); Armstrong et al. (2016):** Warn that the competitive race to develop AGI may result in compromised safety standards and ethical lapses, leading to potentially disastrous consequences.
- **Armstrong et al. (2016); Van de Gevel and Noussair (2013):** Advocate for stringent ethical frameworks and government policies to ensure the safe development of AGI and its benefits to humanity.

The ethics of artificial intelligence: Issues and initiatives

The literature highlights the significant ethical and societal implications of AI, emphasizing the need for robust frameworks and guidelines to address these challenges. Reports underscore the profound potential of AI to transform numerous sectors, offering benefits like improved efficiency and new capabilities. However, the deployment of AI also raises critical ethical questions, including issues of privacy, accountability, bias, and the impact on human relationships. Definitions of AI provide a foundation for examining the moral and ethical considerations involved. National and international strategies emphasize the importance of transparency, fairness, and accountability in AI systems. Additionally, initiatives stress the need for governance frameworks and regulatory bodies to oversee AI development, ensuring that human well-being is prioritized throughout the design phase. These frameworks collectively aim to mitigate risks and ensure that the benefits of AI are distributed equitably, fostering trust and understanding in society.

- **Makridakis (2017):** Highlights AI's potential to transform sectors such as healthcare, transportation, and finance, offering benefits like improved efficiency and new capabilities.
- **Van de Gevel and Noussair (2013):** Discuss critical ethical questions related to AI, including privacy, accountability, bias, and the impact on human relationships.
- **European Commission (2018):** Defines AI as 'systems that display intelligent behaviour', providing a foundation for examining moral and ethical considerations.

- **OECD (2019):** Emphasizes the importance of transparency, fairness, and accountability in AI systems through national and international strategies like the EU Strategy on AI and the OECD Principles on AI.
- **IEEE (2019):** Stresses the need for governance frameworks and regulatory bodies to oversee AI development, ensuring that human well-being is prioritized.

The Ethics of AI Ethics: An Evaluation of Guidelines

The development and dissemination of ethical AI guidelines underscore the critical need for ethical oversight in AI and machine learning advancements. Ethical guidelines serve as essential frameworks to guide developers and researchers in incorporating ethical considerations into AI design and deployment, aiming to mitigate risks and enhance societal benefits. Reports highlight the profound potential of AI to revolutionize various sectors, leading to significant improvements in efficiency and capability. National and international strategies emphasize the importance of transparency, fairness, and accountability in AI systems. Initiatives advocate for governance frameworks and regulatory bodies to oversee AI development, ensuring that human well-being is prioritized throughout the design phase. Additionally, ethical AI guidelines contribute to the promotion of public trust in AI technologies by highlighting the necessity of ethical principles such as privacy, accountability, and the avoidance of bias. These frameworks collectively aim to ensure that the benefits of AI are equitably distributed and that AI systems operate within ethical boundaries, fostering trust and understanding in society.

- **Makridakis (2017):** Highlights AI's potential to revolutionize sectors like healthcare, transportation, and finance, leading to significant improvements in efficiency and capability.
- **OECD (2019):** Emphasizes the importance of transparency, fairness, and accountability in AI systems through strategies like the EU Strategy on AI and the OECD Principles on AI.
- **IEEE (2019):** Advocates for governance frameworks and regulatory bodies to oversee AI development, ensuring human well-being is prioritized.
- **Van de Gevel and Noussair (2013):** Highlights the necessity of ethical principles such as privacy, accountability, and the avoidance of bias in promoting public trust in AI technologies.

11 Common Ethical Issues in Artificial Intelligence

The exploration of legal and ethical considerations in AI technology is supported by a rich body of literature that underscores the complexities and challenges associated with AI deployment. Legal concerns, such as violations of intellectual property rights and data privacy issues, are prominently discussed in scholarly work. Recent legal disputes involving AI highlight significant concerns regarding the unauthorized use of intellectual property. Additionally, data

privacy issues related to AI tools have been critically examined, pointing out potential violations of regulations due to the handling of personal data. On the ethical front, literature addresses a range of issues including job displacement, privacy, and bias. The impact of AI on employment, with automation potentially leading to job losses, has been documented extensively, reflecting widespread concerns about the future of work. Privacy concerns, particularly regarding the use of AI for surveillance and data collection, are also well-documented, emphasizing the need for transparent data practices. Bias in AI systems, driven by biased training data, is another critical issue discussed in the literature, highlighting the risk of perpetuating discrimination through AI technologies. Together, these scholarly insights provide a comprehensive understanding of the legal and ethical challenges posed by AI, underscoring the importance of addressing these issues to ensure responsible AI development and deployment.

- **Pritchard et al. (2023):** Discusses a legal dispute involving OpenAI for alleged copyright infringement, highlighting concerns about unauthorized use of intellectual property.
- **Narayanan et al. (2018); Zuboff (2019):** Examine data privacy issues related to AI tools, pointing out potential violations of regulations like GDPR and CCPA due to personal data handling.
- **Bessen (2019):** Documents the impact of AI on employment, reflecting concerns about job losses due to automation.
- **Andrejevic (2014); Lyon (2018):** Highlight privacy concerns related to AI surveillance and data collection, emphasizing the need for transparent data practices.
- **O'Neil (2016); Barocas et al. (2019):** Address bias in AI systems, driven by biased training data, and the risk of perpetuating discrimination through AI technologies.

CONCLUSION

Artificial Intelligence (AI) is a transformative technology with vast potential to revolutionize various sectors, including healthcare, finance, and education. However, its integration into society comes with significant legal and ethical considerations that must be addressed to ensure responsible and equitable use. The pro-literature evidence highlights the importance of incorporating ethical principles in AI development, emphasizing the need for diverse and representative datasets, informed consent, privacy protection, public involvement, and ethical design. These measures are crucial for minimizing biases, protecting individual rights, and fostering trust and accountability in AI systems.

The con-literature, on the other hand, evidences problems and risks associated with the deployment of AI, including intellectual property violations, data privacy concerns, job displacement, biases that are serious ethical dilemmas within the AI system. Considering the fact that there is a race to develop AI technologies, including Artificial General Intelligence, concerns over the adherence to the safety standards are insignificant as against ethical lapses. Literature calls for robust ethical frameworks, transparent data practices, and stringent government policies to mitigate these risks and promote socially responsible AI.

In conclusion, the responsible development and deployment of AI must be based on a multi-faceted approach that infuses ethical considerations at all stages of the life cycle of AI systems. IT professionals, researchers, industry leaders, and policymakers have to work together in order to evolve comprehensive guidelines and frameworks that would respond to the legal and ethical challenges arising from AI in a complex way. Doing so will allow us the use of power artificial intelligence with protection for human values and rights, thereby fairly using accountable AI technologies that ensure the greater good for societal well-being.

REFERENCES:

Pros:

Li, N. (2023). Ethical Considerations in Artificial Intelligence: A Comprehensive Discussion from the Perspective of Computer Vision. SHS Web of Conferences, 179, 04024.

<https://doi.org/10.1051/shsconf/202317904024>

Bankins, S., & Formosa, P. (2023). The ethical implications of artificial intelligence (AI) for meaningful work. Journal of Business Ethics, 185(4), 725–740.

<https://doi.org/10.1007/s10551-023-05339-7>

Principled artificial intelligence. (2020, January 15). Berkman Klein Center.

<https://cyber.harvard.edu/publication/2020/principled-ai>

Nguyen, A., Ngo, H. N., Hong, Y., Dang, B., & Nguyen, B. T. (2022). Ethical principles for artificial intelligence in education. Education and Information Technologies, 28(4), 4221–4241.

<https://doi.org/10.1007/s10639-022-11316-w>

The role of systems Engineers in ensuring ethical AI practices | MoldStud. (2024, February 4).

<https://moldstud.com/articles/p-the-role-of-systems-engineers-in-ensuring-ethical-ai-practices>

Cons:

Pant, A., Hoda, R., Tantithamthavorn, C., & Turhan, B. (2024). Ethics in AI through the practitioner's view: a grounded theory literature review. *Empirical Software Engineering*, 29(3). <https://doi.org/10.1007/s10664-024-10465-5>

Naudé, W., & Dimitri, N. (2019). The race for an artificial general intelligence: implications for public policy. *AI & Society*, 35(2), 367–379. <https://doi.org/10.1007/s00146-019-00887-x>

The ethics of artificial intelligence: Issues and initiatives | Think Tank | European Parliament. (n.d.). [https://www.europarl.europa.eu/thinktank/en/document/EPRS_STU\(2020\)634452](https://www.europarl.europa.eu/thinktank/en/document/EPRS_STU(2020)634452)

Hagendorff, T. (2020). The Ethics of AI Ethics: An Evaluation of Guidelines. *Minds and Machines*, 30(1), 99–120. <https://doi.org/10.1007/s11023-020-09517-8>

11 Common ethical issues in artificial intelligence. (2023, November 16). Default. <https://connect.comptia.org/blog/common-ethical-issues-in-artificial-intelligence>